ingredients; to most early string researchers, gravity was a distant, secondary concern, at best. By contrast, loop quantum gravity grew out of a tradition tightly grounded in the general theory of relativity; to most practitioners of this approach, gravity has always been the main focus. A one-sentence comparison would hold that string theorists start with the small (quantum theory) and move to embrace the large (gravity), while adherents of loop quantum gravity start with the large (gravity) and move to embrace the small (quantum theory).[9] In fact, as we saw in Chapter 12, string theory was initially developed as a quantum theory of the strong nuclear force operating within atomic nuclei; it was realized only later, serendipitously, that the theory actually included gravity. Loop quantum gravity, on the other hand, takes Einstein's general relativity as its point of departure and seeks to incorporate quantum mechanics.

This starting at opposite ends of the spectrum is mirrored in the ways the two theories have so far developed. To some extent, the main achievements of each prove to be the failings of the other. For example, string theory merges all forces and all matter, including gravity (a complete unification that eludes the loop approach), by describing everything in the language of vibrating strings. The particle of gravity, the graviton, is but one particular string vibrational pattern, and hence the theory naturally describes how these elemental bundles of gravity move and interact quantum mechanically. However, as just noted, the main failing of current formulations of string theory is that they presuppose a background spacetime within which strings move and vibrate. By contrast, the main achievement of loop quantum gravity—an impressive one—is that it does *not* assume a background spacetime. Loop quantum gravity is a background-independent framework. However, extracting ordinary space and time, as well as the familiar and successful features of general relativity when applied on large distance scales (something easily done with current formulations of string theory) from this extraordinarily unfamiliar spaceless/timeless starting point, is a far from trivial problem, which researchers are still trying to solve. Moreover, in comparison to string theory, loop quantum gravity has made far less progress in understanding the dynamics of gravitons.

One harmonious possibility is that string enthusiasts and loop quantum gravity aficionados are actually constructing the same theory, but from vastly different starting points. That each theory involves loops—in